

Supervised Machine Learning

QSS20: Modern Statistical Computing

Looking back! What's the point of all this Python?

- **Dataframe manipulation**, aggregation, merging, etc
 - Using **pandas**
 - Writing **functions** for general purpose computing
- **Visualizing** these results using Matplotlib or ggplot

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 - Using regular expressions, POS tagging, NER to **manipulate text**
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Building quantitative models via regression or categorization

Farm to Table



Harvesting

Raw Data

- APIs
- `pd.read_csv`



Processing

Data Augmentation

- LDA Topics
- Sentiment



Transportation

Merging Datasets



Dinner

Visualization

Farm to Table



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Transporation

Merging Datasets



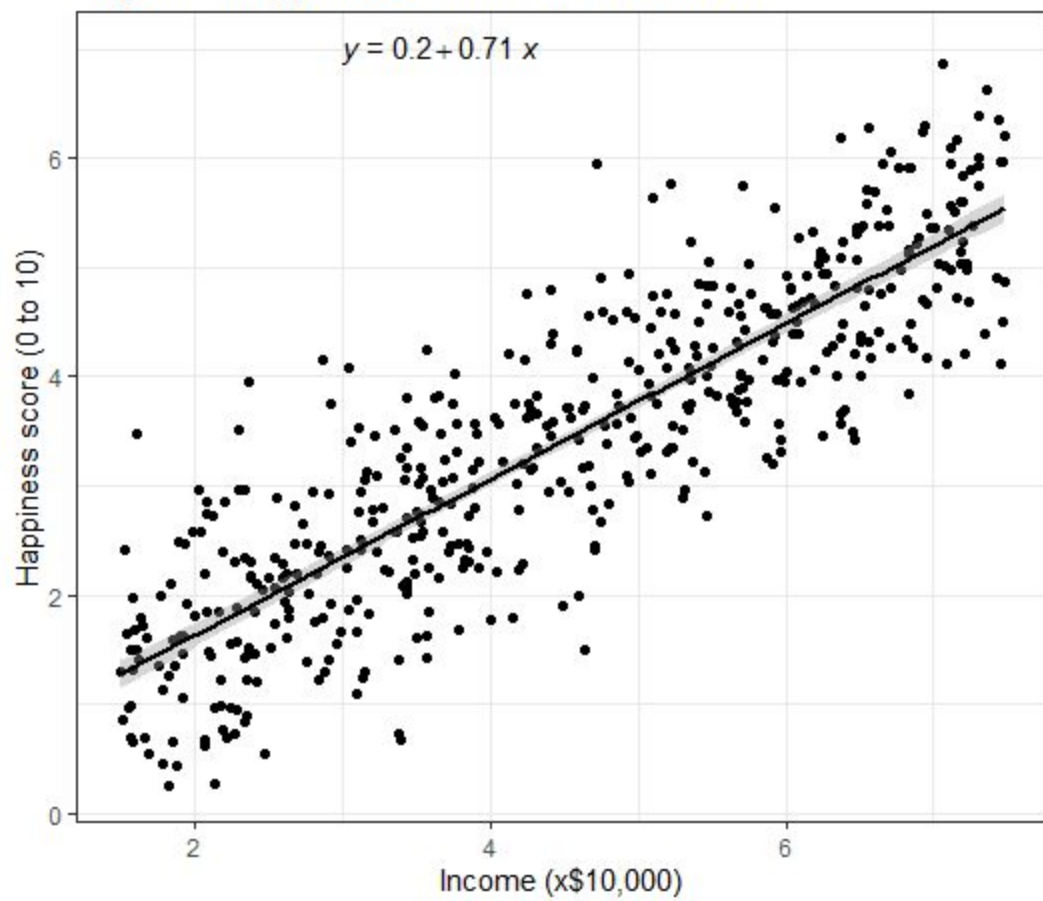
Regression



Dinner

Visualization

Reported happiness as a function of income



What is regression?

"Regression to the mean"

Galton observed...

- Tall parents tended to have shorter children (relatively)
- Shorter parents tended to have taller children (relatively)
- Children's heights averaged out more toward the population mean.



Two types of regression

- Classification
- Regression

Spam / Not Spam ?

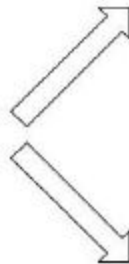
- *Hey @dancer317, love ur videos so much! Thanks for all the tips on dancing!*
- *OUR LASER PRINTER/FAX/COPIER TONER CARTRIDGE PRICES NOW AS LOW AS 39 DOLLARS. SPECIALS WEEKLY ON ALL LASER PRINTER SUPPLIES. WE CARRY MOST ALL LASER PRINTER CARTRIDGES, FAX SUPPLIES AND COPIER TONERS AT WAREHOUSE PRICES*
- *I'm not sold on your first point about crossing national boundaries, but I see what you mean about non-economic alternatives.*

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*Judge first floor is where I met your grace
A Manhattan girl from downtown with a beautiful face
We could ski together anyplace
And then sit down next to a fireplace*

Hey @dancer317,
love ur videos so
much! Thanks for all
the tips on dancing!

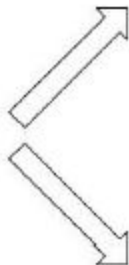


Spam

Not spam

We want our machines to learn to do this, like so:

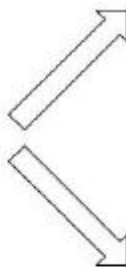
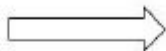
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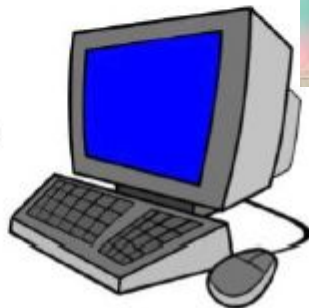
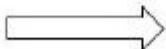
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Spam

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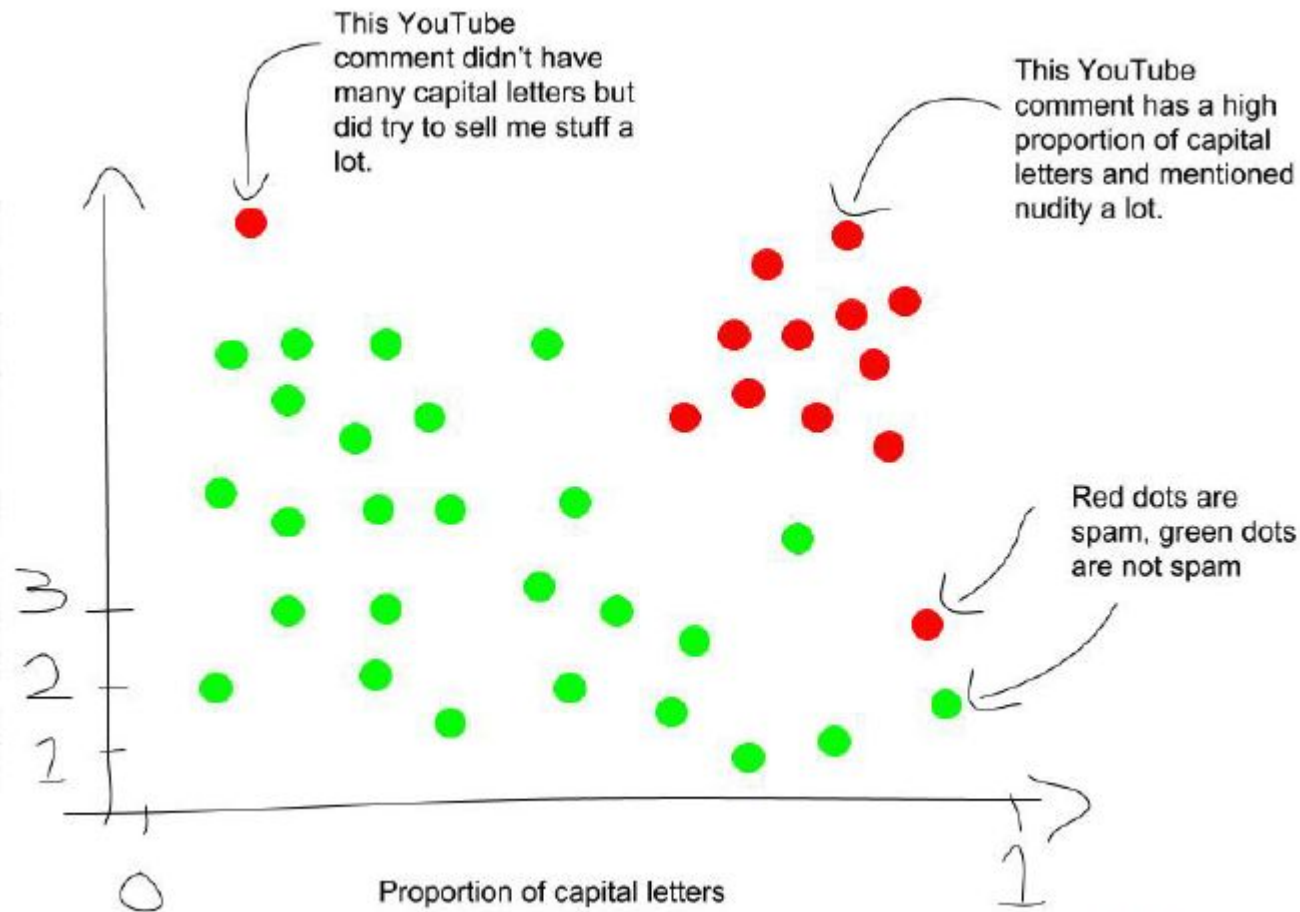
Not spam

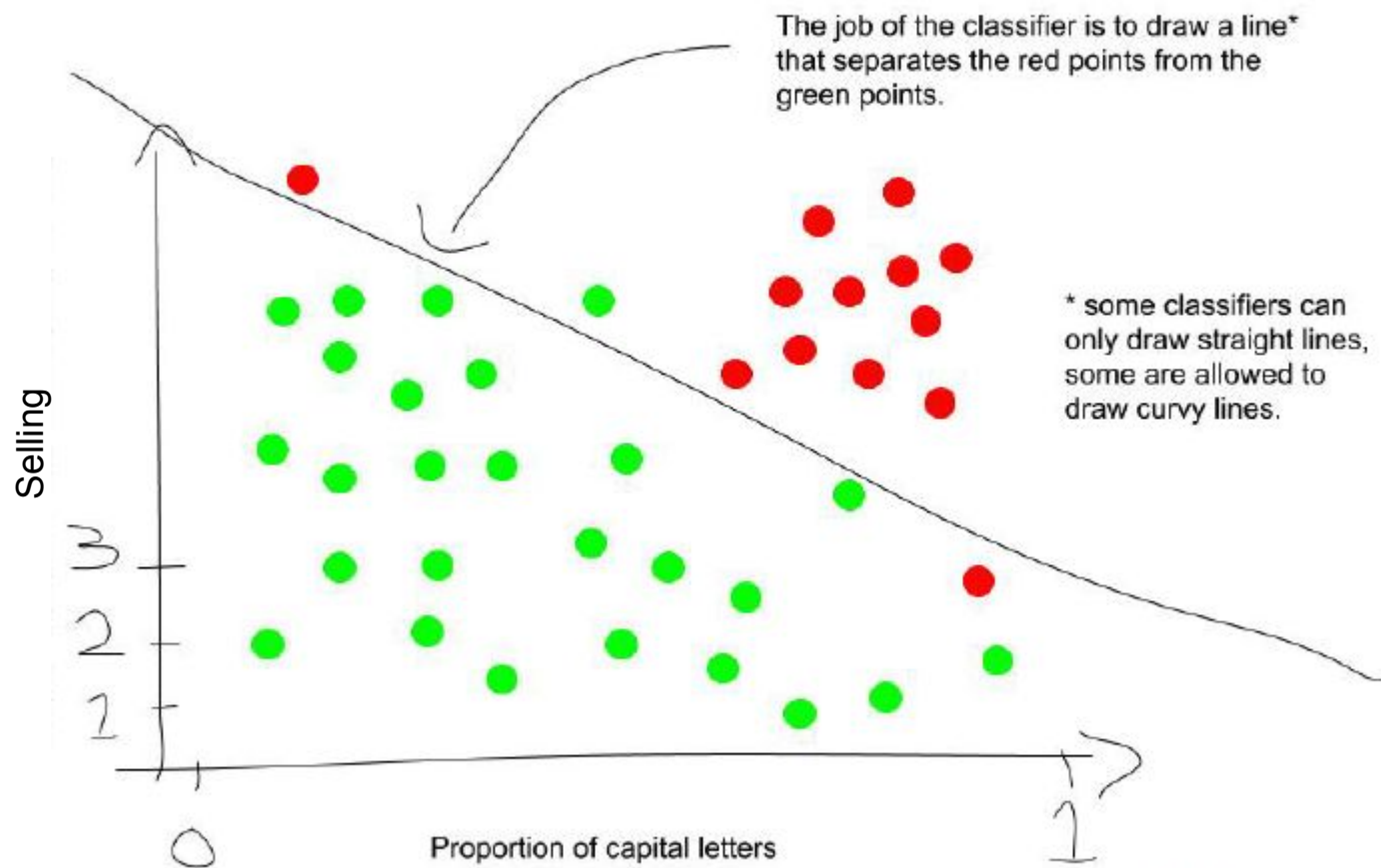
I appreciate your thought
And I am glad that you shot your shot
But I only wish to be friends
I hope that does not offend

How computers see data

Comment	Selling	Prop. of capital letters	Spam?
Hey @dancer,...	0	0.1	No
OUR LASER PRINTER....	0.8	1.0	Yes
I'm not sold on your first point...	0.5	0.05	No
Judge first floor is...	0	0.05	No

Selling





Where have we seen machine learning?

Where have we seen machine learning?

- **Sentiment classification?**

Uses a dictionary approach so not technically machine learning

- **Topic Modeling**

LDA uses unsupervised machine learning that takes unlabeled input.

Spits out a bunch of probability topics and words.

Selected Topic: 1

Previous Topic

Next Topic

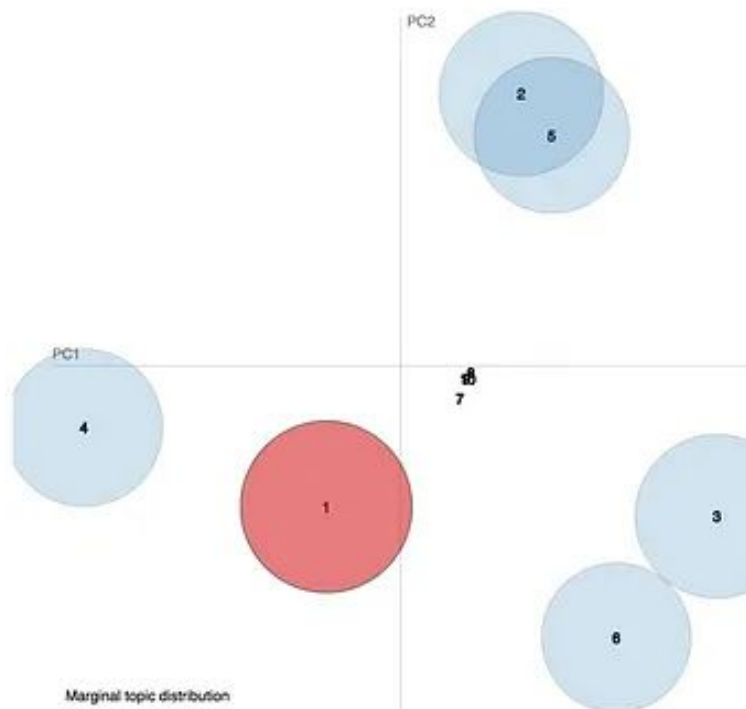
Clear Topic

Slide to adjust relevance metric:(2)

 $\lambda = 0.29$

0.0 0.2 0.4 0.6 0.8 1.0

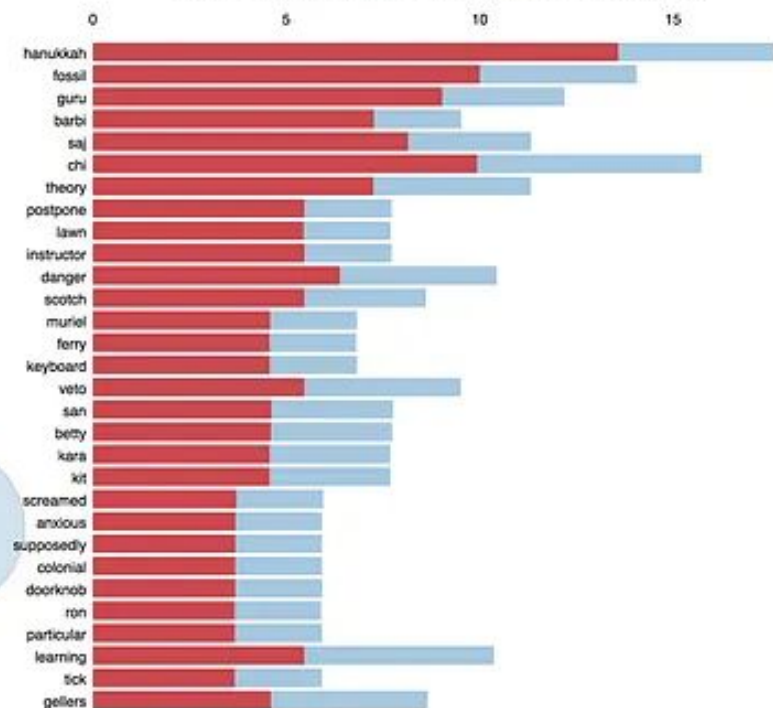
Intertopic Distance Map (via multidimensional scaling)



Marginal topic distribution



Top-30 Most Relevant Terms for Topic 1 (19.1% of tokens)



Overall term frequency

Estimated term frequency within the selected topic

1. saliency(term w) = frequency(w) * $\{\sum_i p(t | w) * \log(p(t | w)/p(t))\}$ for topics t ; see Chuang et. al (2012)2. relevance(term w | topic t) = $\lambda * p(w | t) + (1 - \lambda) * p(w | t)/p(w)$; see Sievert & Shirley (2014)

Two flavors of supervised ML

1. **Regression:** Predict numeric values.
2. **Classification:** Predicting categories.
 - a. Binary classification: Predicting yes or no.
 - b. Multiclass Prediction: Predicting one or more classes.

Why do supervised?

- After labeling 300 or 400 cases, let the computer do it for hundreds of thousands. Scales your results.

Example of binary classifiers gone wrong

<https://www.youtube.com/watch?v=ACmydtFDTGs>

Terminology

Labeled data

- Target variable in computer science
- Dependent variable in social science
- Examples: Spam/not spam, voting yes or no

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Features

- Covariates or independent variables—things used to predict the dependent
- Text: Parts of speech, sentiment, etc.

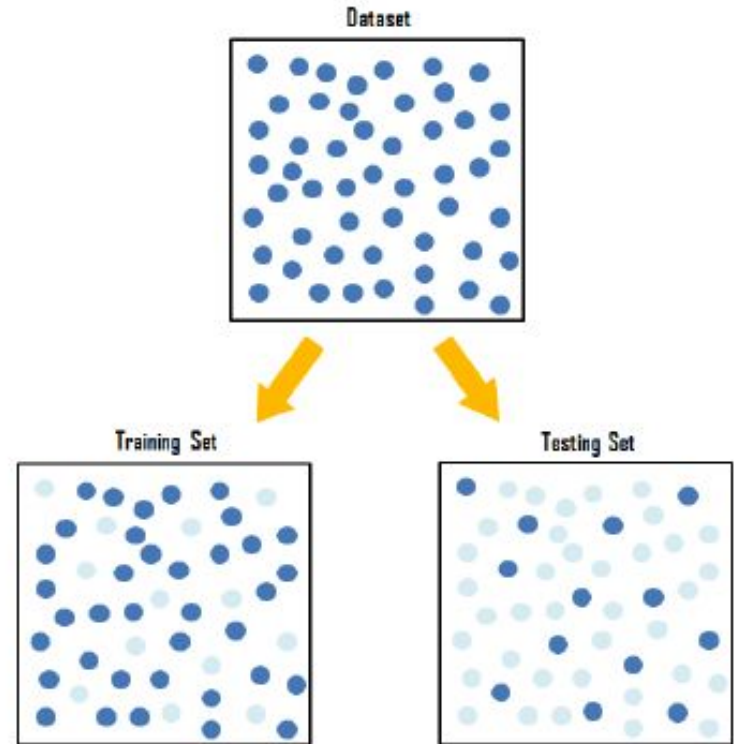
One thing you should take away from QSS20

[illegible]

ML in Theory

Key task for machine learning: Train-test split

- Split the data into two parts.
 - 80% for training.
 - 20% for testing.
- You train your model on the 80% and then let it loose on the 20%.



What's the point? One thing you should take away

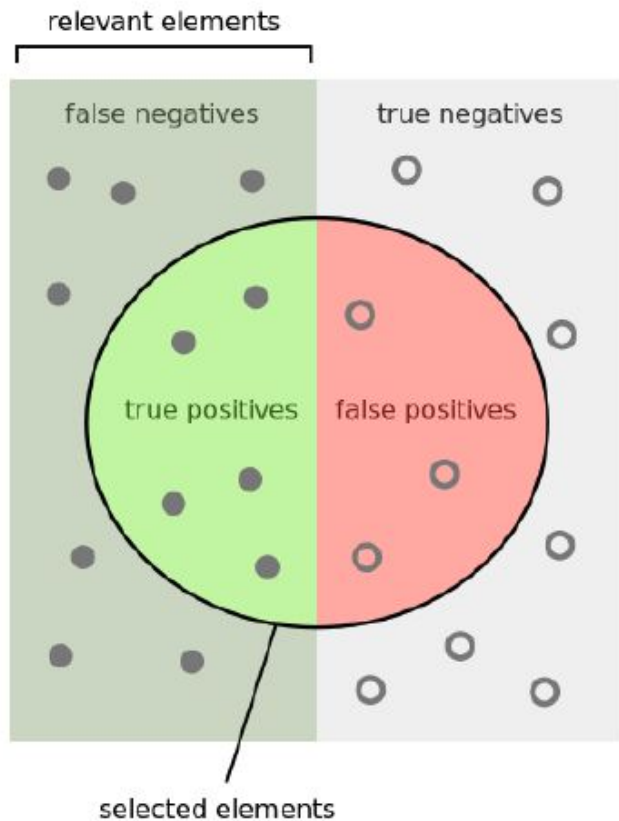
	Independent Variables, features, covariates					Label or value
	x1	x2	x3	x4	x5	y
Training						
Testing						

How do we evaluate these models? Confusion matrix

		Predicted results	
		True	False
Actual Results	True	True positives	False Negative
	False	False Positive	True negative

Confusion matrices are one of the most useful tables you can create.
Also for overlap between two categories.

Precision and recall



How many selected items are relevant?

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

How many relevant items are selected?

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Results}}$$

$$\text{Recall} = \frac{\text{True Positives}}{\text{All Positives}}$$

All the key metrics

Accuracy:

$$\frac{\sum \text{true positives} + \sum \text{true negatives}}{\sum \text{total population}}$$

Precision:

$$\frac{\sum \text{true positives}}{\sum \text{predicted positives}}$$

Recall:

$$\frac{\sum \text{true positives}}{\sum \text{condition positives}}$$

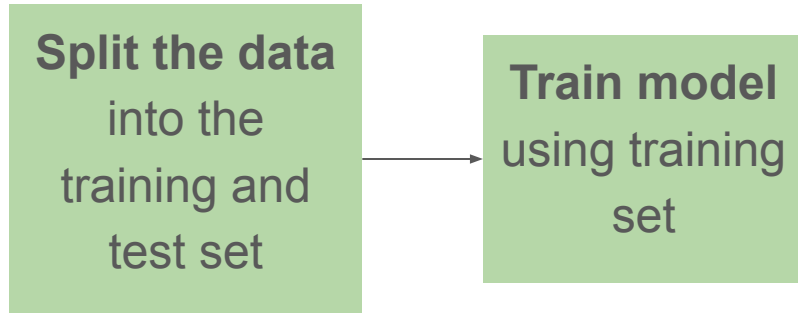
F1-score:

$$\frac{\sum \text{true positives}}{\sum \text{true positives} + 1/2(\sum \text{false positives} + \sum \text{false negatives})}$$

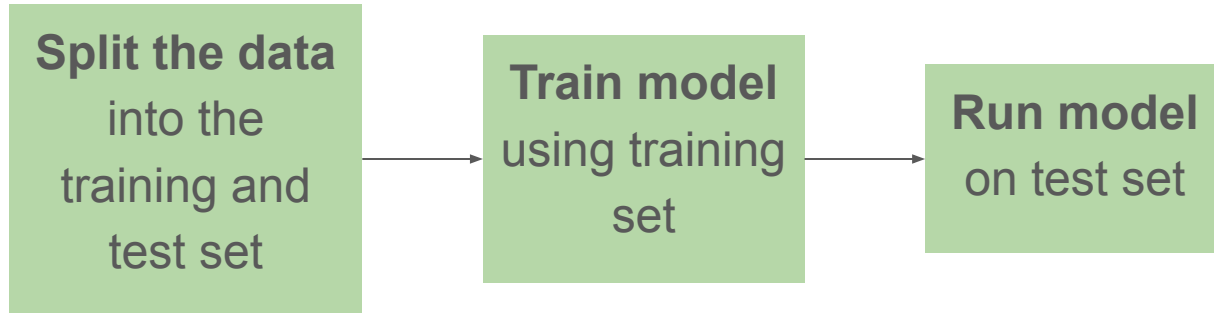
Stages of ML

Split the data
into the
training and
test set

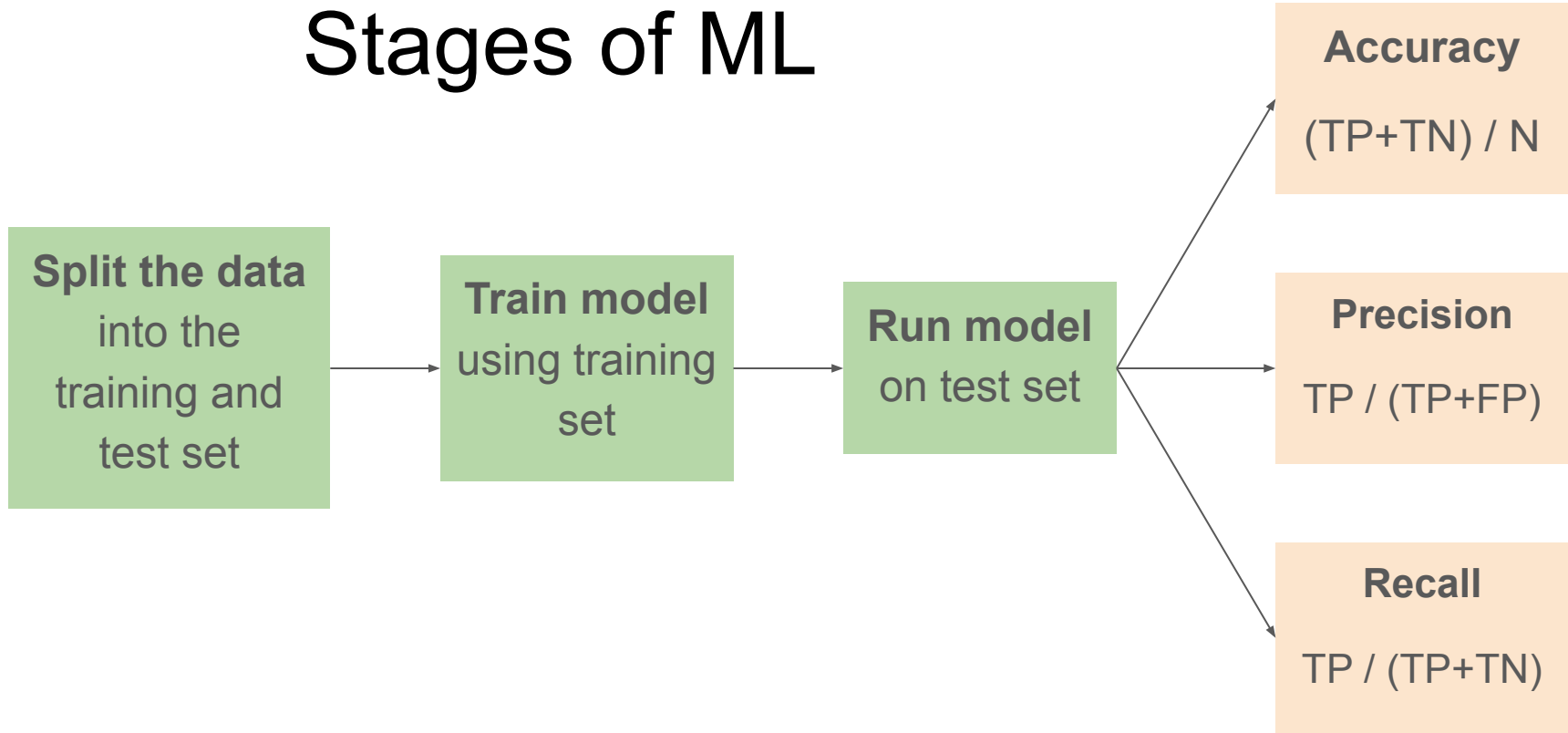
Stages of ML



Stages of ML



Stages of ML



What is the model? Take in X spit out Y.

- Linear regression
- Logistic regression
- Decision trees
- Gradient-boosting trees
 - XGBoost
 - CatBoost
- Deep-learning

Simple and explainable
(explained variance)

Complex and black box

Two types of outcome variables

- Continuous
- Categorical

Two types of outcome variables

- **Continuous**
- Categorical

Linear Model

All predictions are a linear combination of each variable

The diagram illustrates the components of a linear model equation. The equation is $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \varepsilon$. Labels with arrows point to specific parts of the equation: 'Dependent Variable (Response Variable)' points to Y ; 'Independent Variables (Predictors)' points to the X terms; 'Y intercept' points to β_0 ; 'Slope Coefficient' points to β_1 and β_2 ; and 'Error Term' points to ε .

Dependent Variable
(Response Variable)

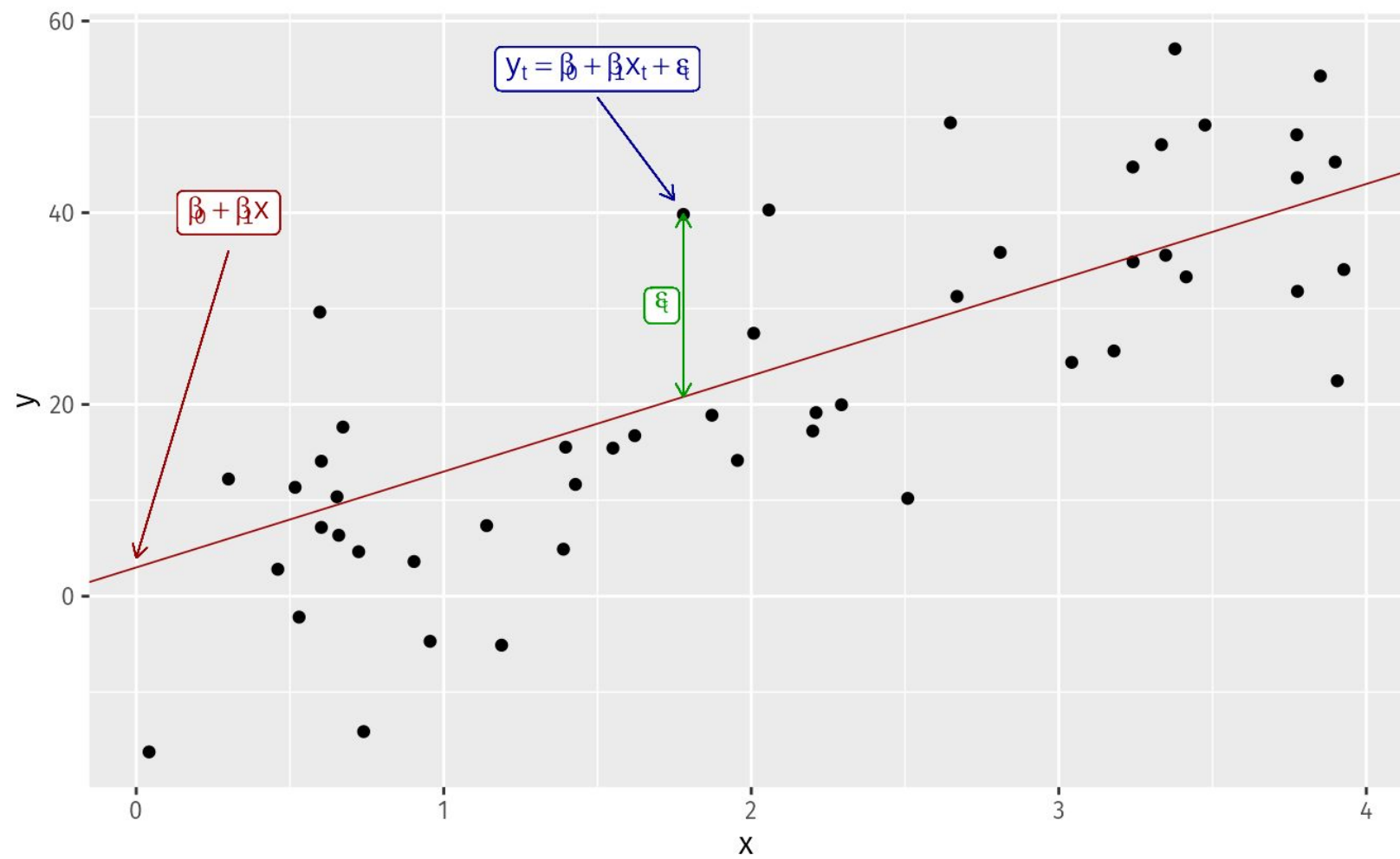
Independent Variables
(Predictors)

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \varepsilon$$

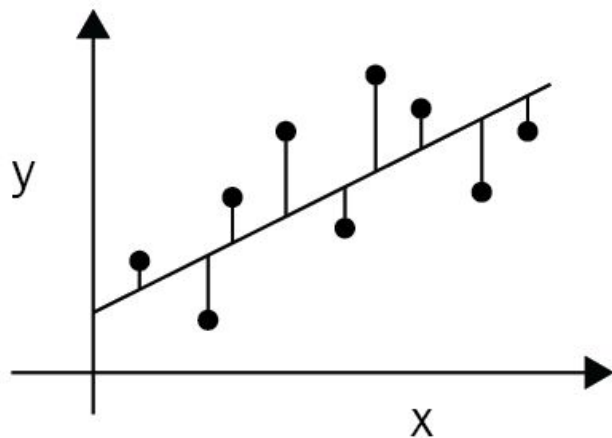
Y intercept

Slope
Coefficient

Error Term



How do we measure error?

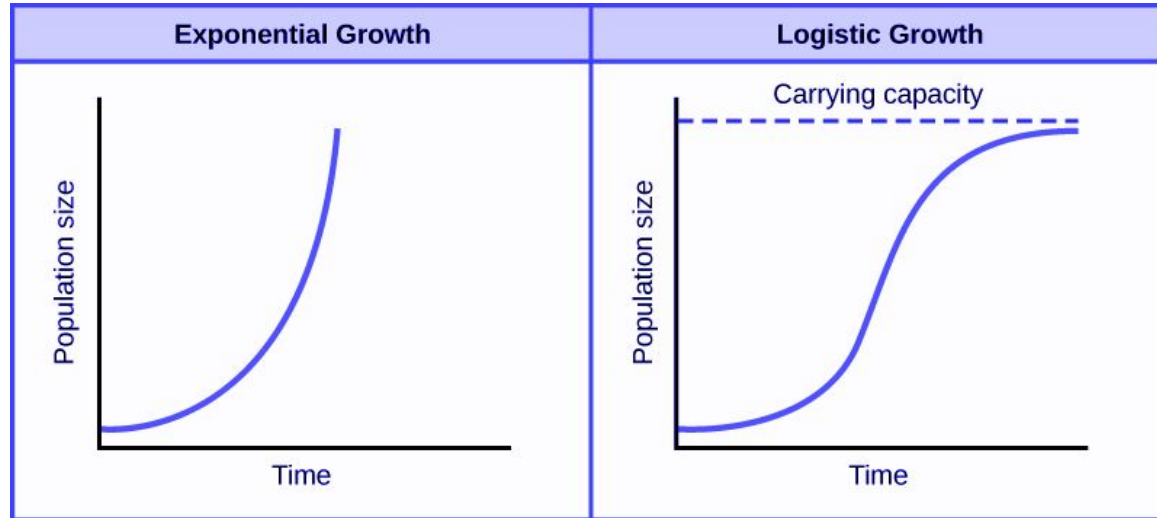


$$SS_{residuals} = \sum_{i=1}^N (\hat{y}_i - y_i)^2$$

Model Prediction
↓
Observed Result

The logistic function | Sigmoid function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$



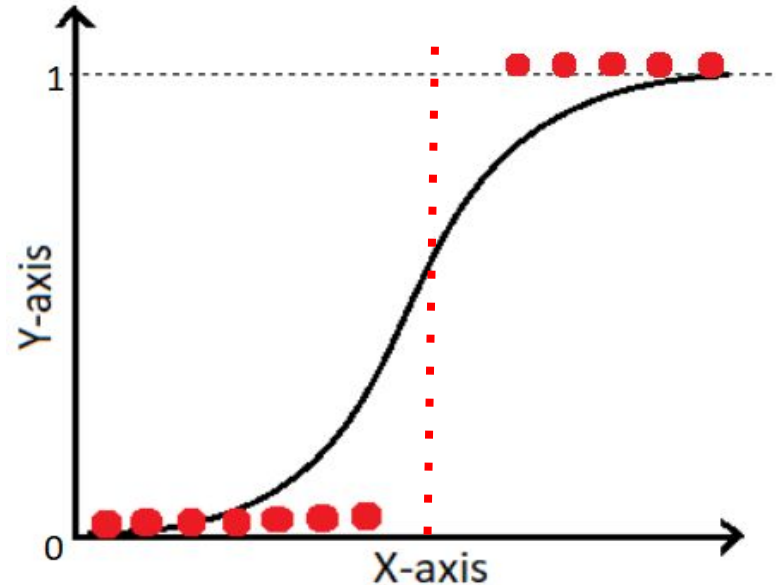
General form of logistic regression

$$P(Y = 1 \mid X_1, X_2, \dots, X_p) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p)}}$$

Logistic regression (LOGIT)

- Probability of positive label is described by a sigmoid curve.
- Note the interior expression is a **linear function**.
- Thus, when the function is greater than 0.5, then this maps to a positive label.

$$P = \frac{e^{a+bX}}{1 + e^{a+bX}}$$



Flavors of models (don't worry too much!)

L1 regularization:

Lasso Regression

Minimizes on the absolute value of coefficients

Used to reduce complex models to essential params

L2 regularization:

Ridge Regression

Minimizes on the square of coefficients

MISC | Distance Metrics

L1 Regularization:

$$\sum_{i=1}^n (Y_i - \sum_{j=1}^p X_{ij} \beta_j)^2 + \lambda \sum_{j=1}^p |\beta_j|$$

L2 Regularization:

$$\sum_{i=1}^n (y_i - \sum_{j=1}^p x_{ij} \beta_j)^2 + \lambda \sum_{j=1}^p \beta_j^2$$

Some intuitions

- L1: The less coefficients you have, the better
- L2: The smaller coefficients you have, the better

ML in Python

Example with Yelp | 1 = positive sentiment

metadata_label

raw_text

0 Unfortunately, the frustration of being Dr. Goldberg's patient is a repeat of the experience I've had with so many other doctors in NYC -- good doctor, terrible staff. It seems that his staff simply never answers the phone. It usually takes 2 hours of repeated calling to get an answer. Who has time for that or wants to deal with it? I have run into this problem with many other doctors and I just don't get it. You have office workers, you have patients with medical needs, why isn't anyone answering the phone? It's incomprehensible and not work the aggravation. It's with regret that I feel that I have to give Dr. Goldberg 2 stars.

1 Been going to Dr. Goldberg for over 10 years. I think I was one of his 1st patients when he started at MHMG. He's been great over the years and is really all about the big picture. It is because of him, not my now former gyn Dr. Markoff, that I found out I have fibroids. He explores all options with you and is very patient and understanding. He doesn't judge and asks all the right questions. Very thorough and wants to be kept in the loop on every aspect of your medical health and your life.

0 I don't know what Dr. Goldberg was like before moving to Arizona, but let me tell you, STAY AWAY from this doctor and this office. I was going to Dr. Johnson before he left and Goldberg took over when Johnson left. He is not a caring doctor. He is only interested in the co-pay and having you come in for medication refills every month. He will not give refills and could less about patients's financial situations. Trying to get your 90 days mail away pharmacy prescriptions through this guy is a joke. And to make matters even worse, his office staff is incompetent. 90% of the time when you call the office, they'll put you through to a voice mail, that NO ONE ever answers or returns your call. Both my adult

Uses word “frustrat” and negative label:

metadata_label		raw_text	frustrat
0	Say \"BITES.\" Say \"WINGS.\" Apparently, my waiter did not know the difference between \"WINGS\" and \"BITES,\" because when my roommate and I ordered buffalo BITES, we got WINGS instead. We hate wings and love bonless buffalo BITES. It was even more frustrating when our buddy came to join us at happy hour and the waiter spoke for two minutes about how he prefers the BITES over the WINGS. I know I probably sound dramatic, but I will NOT be back...		1
0	This used to be my \"go to\" place for groceries. However, lately I'm becoming more and more frustrated by the store. If you go between 4pm and 6pm on a weekday or at any time on Saturday or Sunday, be prepared to navigate a mess of a parking lot, an over crowded store, and a 20 minute wait in a checkout lane. On a Sunday, it's impossible to find parking even though there is outdoor parking as well as a 2 story garage. Once you get in the store, you have to navigate through an obstacle course of a produce section and around overwhelmed first-timers who are trying to figure out where everything is. It's entirely too small for a place that markets itself towards the \"fresh foods\" end of things (as opposed to a regular Giant Eagle). Then you have to figure out where everything is in the aisles. I get the impression that things were just randomly arranged around the store. Once you actually have your groceries, the waiting begins.		1

Step 2: Split the data into train and test sets

```
## define the number of rows in training set (20\% test)
## and sample
nrows_train = round(X.shape[0]*0.8)
nrows_test = X.shape[0] - nrows_train
random.seed(221)
train_ids = random.sample(set(X['metadata_rowid']),
                           nrows_train)
## function fed ids and name of id col
def my_split(train_ids, id_col):
    test_ids = set(X[id_col]).difference(train_ids)
    X_train_man = X[X[id_col].isin(train_ids)].copy()
    X_test_man = X[X[id_col].isin(test_ids)].copy()
    y_train_man = y[y.index.isin(train_ids)].iloc[:,
                                                    0].to_numpy()
    y_test_man = y[y.index.isin(test_ids)].iloc[:,
                                                  0].to_numpy()
    return(X_train_man, X_test_man, y_train_man, y_test_man)
```

Step 2: Split the data into train and test sets

[illegible]

Step 3: Estimate the model with training data

Columns we want to exclude

```
## list of nonfeatures
non_feat = ['metadata_rowid', 'raw_text', 'process_text']
## initialize classifier
logit_lasso = LogisticRegression(penalty = "l1",
                                max_iter=100, C = 0.01, solver='liblinear')
## estimate the model on training data/training label
logit_lasso.fit(X_train_man[[col for col in
                             X_train.columns if col not in
                             non_feat]], y_train_man)
```

Create the
logit model

Fit the data to the model

Step 4: Apply model to the test set for predictions

Binary

```
y_pred = logit_lasso.predict(X_test_man[[col for col
                                         in X_test_man.columns if "metadata_rowid"
                                         not in col]])

y_predprob = logit_lasso.predict_proba(X_test_man[[col for col
                                                    in X_test_man.columns if "metadata_rowid"
                                                    not in col]])
```

Prob.

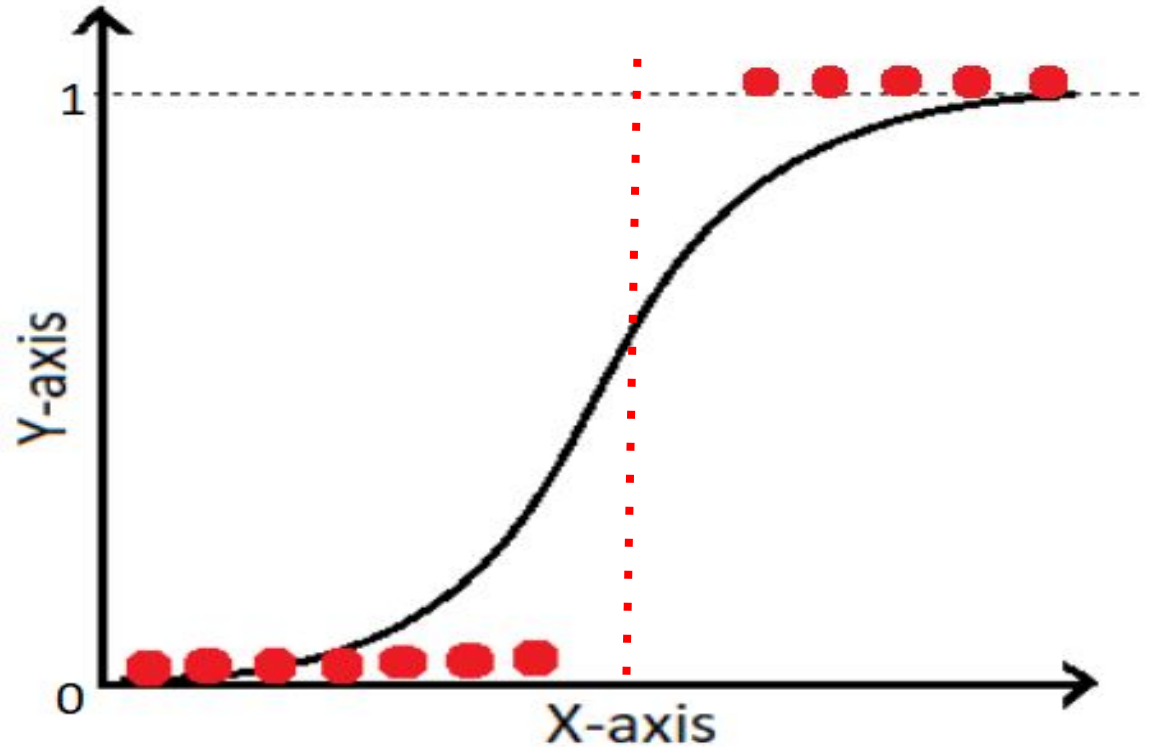
```
## print the results
y_pred[0:5]
y_predprob[0:5]
```

```
array([0, 0, 1, 0, 0])
```

```
array([[0.70544662, 0.29455338],
       [0.70544662, 0.29455338],
       [0.1918429 , 0.8081571 ],
       [0.70544662, 0.29455338],
       [0.70544662, 0.29455338]])
```


Logistic regression (LOGIT)

$$P = \frac{e^{a+bX}}{1 + e^{a+bX}}$$



Step 5: Use prediction to categorize your results (accuracy)

```
## dataframe of binary, continuous, and true labels
y_pred_df = pd.DataFrame({'y_pred_binary': y_pred,
                           'y_pred_continuous':
                               [one_prob[1] for one_prob in y_predprob],
                           'y_true': y_test_man})

## use np.select to code each obs to its error cat
error_cond = [ (y_pred_df['y_true'] == 1) &
                (y_pred_df['y_pred_binary'] == 1),
                (y_pred_df['y_true'] == 1) &
                (y_pred_df['y_pred_binary'] == 0),
                (y_pred_df['y_true'] == 0) &
                (y_pred_df['y_pred_binary'] == 0)]
error_codeto = ["TP", "FN", "TN"]

y_pred_df['error_cat'] = np.select(error_cond,
                                    error_codeto, default = "FP")

## see notebook for calculations using the error_cat column
```


Step 6: Interpret the model

```
feat_imp = pd.DataFrame({'coef': logit_lasso.coef_[0],  
                          'feature_name':  
                          [col for col in X_train.columns  
                           if col not in non_feat]}))
```

- Lists all the coefficients and the words– see what are the most important.

Notebook for today

https://github.com/herbertfreeze/QSS20-X26/blob/main/activities/08_ML_intro_activity_blank.ipynb